



External exposure, boundary-spanning, and opinion leadership in remote communities: A network experiment

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ABSTRACT

Are boundary spanners opinion leaders in ethnically segregated remote low-income communities or are they shunned? Can external exposure create opinion leaders in such peripheral communities? To answer these questions, we invited randomly selected farmers from 16 randomly selected communities in Sumatra to three-day networking and training events outside of their villages. The substantive purpose of these events was for the farmers to learn new practices from their peers in the visited locations. Eighteen months later, we conducted a sociocentric survey of information-sharing networks in the 16 communities. These 16 networks included 380 members, of which 117 participated in our randomized intervention and 263 were in the control group. We found that participants of our randomized intervention had an average indegree that was double that of the control group (2.8 vs 1.4). We applied Exponential Random Graph Models to the 16 networks to account for endogenous network tendencies. We treated participation in the intervention and the number of boundary-spanning links of each actor as node covariates. Results from our models show that actors who participated in the intervention had higher levels of influence in their communities than the control group, and actors with more boundary spanning links were more popular sources of advice. The results suggest that network interventions do not always need to rely on opinion leaders. Under certain conditions, interventions can create opinion leaders by changing local social networks. We conclude with methodological implications for using interventions in social network research.

1. Introduction

Social networks are important channels for learning, innovation, and information diffusion in isolated agrarian communities (Isaac et al., 2014, 2007; Bodin and Prell, 2011). For the inhabitants of such communities, direct exposure to the external world can be an eye-opening experience. Meeting people outside of their village can open access to knowledge that is unavailable at home (Matouš and Todo, 2018). However, traditional village dwellers might not always see the value of such exposure, and information coming from external sources might not be widely accepted, particularly in societies characterised by high levels of territorialism or ethnic fragmentation (Barnes et al., 2016).

This study is about boundary spanning and opinion leadership in remote communities. Boundary spanning and opinion leadership are distinct but potentially intertwined concepts. Boundary spanners communicate across the boundaries of their groups (Shah et al., 2018). Opinion leaders are popular individuals whom others seek for information and whose practices are likely to be imitated (Valente and

Davis, 1999; Parau et al., 2017).

In peripheral communities, new ideas and practices often come from outside and are adopted first by those on the margin of the local networks whose attention is directed outwards (Valente, 1995). New practices diffuse widely in the local communities only if opinion leaders accept them and opinion leaders will be careful to do that. If external influences are generally frowned upon, opinion leaders may be reluctant to engage with the external world and adopt practices coming from outside. In contrast, if boundary spanning brings respect and prestige, the people whose links cross the network boundary will be in the centre of community attention, and the diffusion of external innovations will become faster (Rogers, 2003).

Understanding these mechanisms and the ways to manipulate them is useful for disseminating information in isolated settlements, promoting good practices, and designing network interventions that could effectively improve people's lives in marginalized communities. Numerous studies have been conducted on personal attributes of opinion leaders (Rogers, 2003) but we still do not sufficiently understand

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